데이터베이스 501분반

캠퍼스 활동 한눈에, 공간 예약 한 번에

CAMPUS LINK

Team 7 | 김다경, 문다민, 정재영, 최준태

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01 Background

→ **기존 앱의 기능 부족으로 인해**
학생들의 교내 시설 이용 및 활동 정보 접근에 어려움이 발생함

02 Motivation

교내 시설 예약의 한계

기존 교내 앱에는 스터디룸 및 체육시설 (축구장·풋살장) 예약 기능이 부재
학생들은 오프라인 문의 또는 비공식 경로에 의존
이로 인해 이용 가능 여부 확인과 예약 과정에 제약 발생

동아리·활동 정보 접근성 부족

교내 동아리 목록 및 세부 활동 정보를 한눈에 확인할 수 있는 채널 부재
활동 내용, 모집 여부, 운영 방식 등을 파악하기 어려움
신입생 및 재학생 모두 활동 탐색 과정에서 정보 격차 발생

반복되는 학생 불편

시설 이용과 활동 정보 확인 과정이 비효율적이고 분산됨
학생들은 매번 직접 찾아보거나 문의해야 하는 구조
결과적으로 캠퍼스 활동 참여 진입 장벽 지속

02 Solution Direction

첫 번째 포인트!

구조화된 DB를 기반으로

→ 스터디룸·체육시설 예약 프로세스를 전산화

→ 실시간 이용 현황 조회 및 간편 예약 환경 제공

두 번째 포인트!

카테고리화된 DB 설계를 통해

→ 동아리 및 교내 활동 정보를 체계적으로 관리

→ 학생들이 필요한 정보를 쉽게 탐색하고 접근할 수 있는 환경 구축

02 Team Members

김다경 C284005	문다민 C187010	정재영 c384042	최준태 C284047
• 조장 • USERS 설계 • ACTIVITY 설계 • FRONT-END 개발 • 활동/카테고리 BACK-END 개발	• SPACE 설계 • MYPAGE 설계 • FRONT-END 개발 • 공간예약 BACK-END 개발	• CALENDAR 설계 • INDEX 설계 • FRONT-END 개발 • 일정관리 BACK-END 개발	• COMMUNITY설계 • BOARD 설계 • FRONT-END 개발 • 커뮤니티 BACK-END 개발

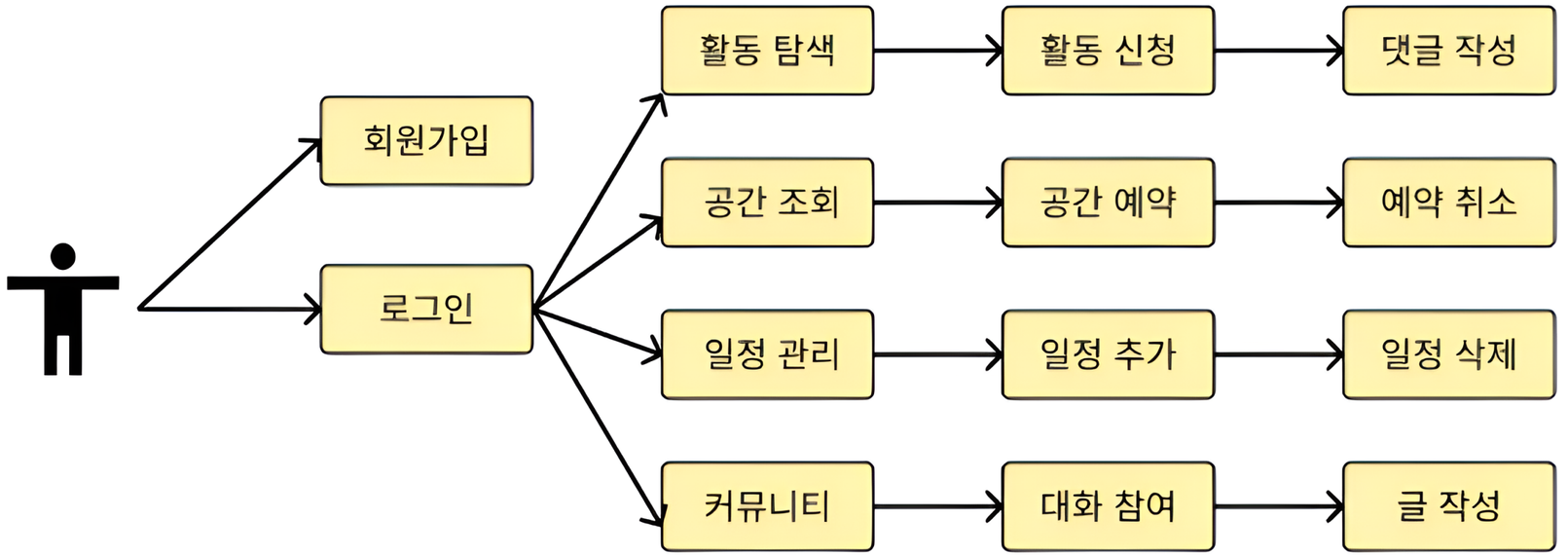
03 SRS

구분	기능명	내용	중요도
공통	회원가입	아이디/ 비밀번호/ 연락처 등을 압력해 회원가입	중
	로그인	아이디/ 비밀번호를 입력해 로그인	중
학생	활동 추천	댓글이 많은 활동 게시글을 바탕으로 활동 추천	상
	활동 관리	활동에 대한 신청 -> 신청 내역은 마이페이지에서 조회	하
		활동 후 학생들은 댓글 작성 -> 수정 및 삭제 가능	중

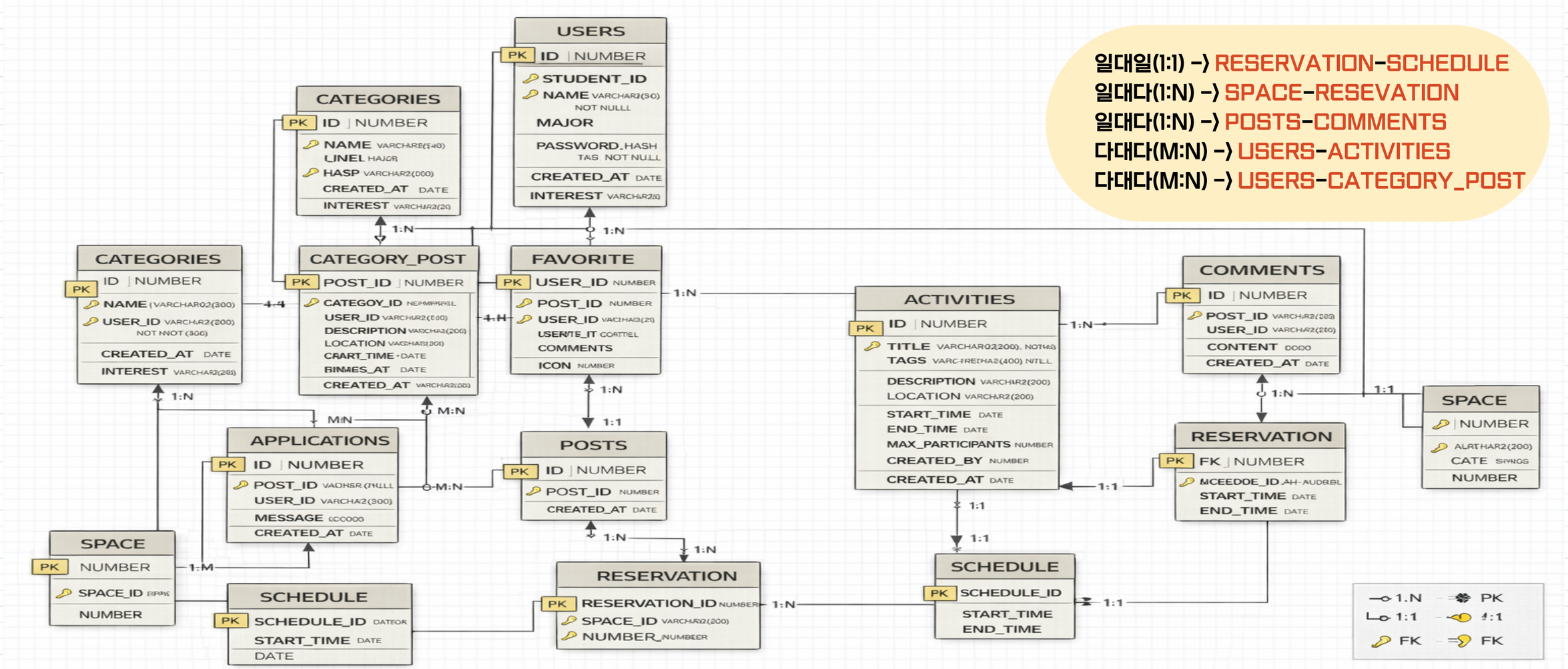
03 SRS

구분	기능명	내용	중요도
학생	일정 연동	개인 일정 등록 및 공간예약 일정 연동	상
		일정 충돌 여부를 자동 확인	중
	공간 예약	강의실 및 공부방 등 예약 기능 제공	중
		예약 가능 시간 및 수용 인원 확인 기능 제공	상
		예약 변경 및 취소 기능 제공	중
	커뮤티네이션	활동별 채팅방을 제공하여 참여간 소통 가능	상

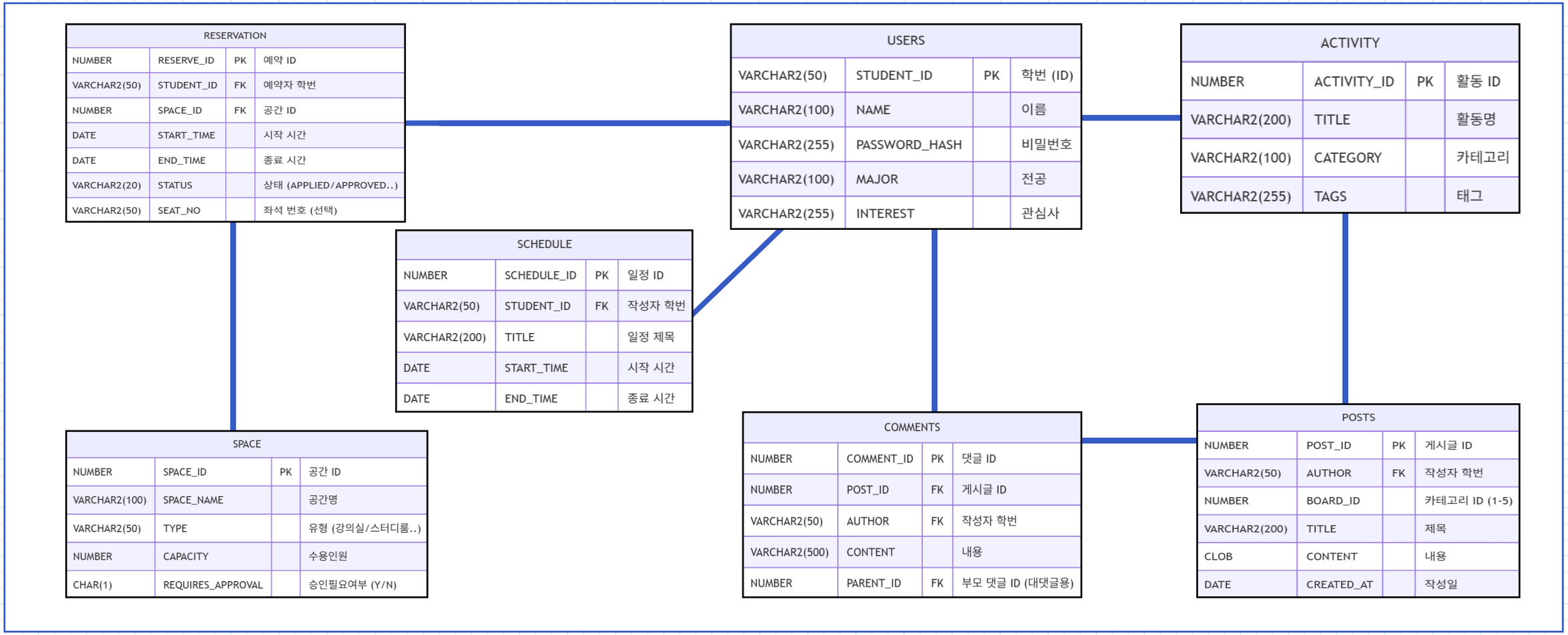
04 Use Case Diagram



04 E-R 다이어그램


일대일(1:1) -> RESERVATION-SCHEDULE
일대다(1:N) -> SPACE-RESEVATION
일대다(1:N) -> POSTS-COMMENTS
다대다(M:N) -> USERS-ACTIVITIES
다대다(M:N) -> USERS-CATEGORY_POST
USERS
PK ID | NUMBER
STUDENT_ID
NAME VARCHAR2(50) NOT NULLL
MAJOR
PASSWORD_HASH TAS NOT NULLL
CREATED_AT DATE
INTEREST VARCHAR2(20)
CATEGORIES
PK ID | NUMBER
NAME VARCHAR2(50)
HASP VARCHAR2(200)
CREATED_AT DATE
INTEREST VARCHAR2(20)
CATEGORIES
PK ID | NUMBER
NAME (VARCHAR02(300)
USER_ID VARCHAR2(200) NOT NNOT (306)
CREATED_AT DATE
INTEREST VARCHAR2(20)
CATEGORY_POST
PK POST_ID | NUMBER
CATEGOY_ID
USER_ID VARCHAR2(100)
DESCRIPTION VARCHAR2(200)
LOCATION VARCHAR2(20)
CRART_TIME DATE
BINMES_AT DATE
CREATED_AT VARCHAR2(20)
FAVORITE
PK USER_ID NUMBER
POST_ID NUMBER
USER_ID VACHAR2(21)
USERNTE_IT CORTTEL
COMMENTS
ICON NUMBER
ACTIVITIES
PK ID | NUMBER
TITLE VARCHAR02(200), NOT40
TAGS VARCHARCHAR2(400) NTLL
DESCRIPTION VARCHAR2(200)
LOCATION VARCHAR2(200)
START_TIME DATE
END_TIME DATE
MAX_PARTICIPANTS NUMBER
CREATED_BY NUMBER
CREATED_AT DATE
COMMENTS
PK ID | NUMBER
POST_ID VARCHAR2(200)
USER_ID VARCHAR2(200)
CONTENT DOOO
CREATED_AT DATE
SPACE
NUMBER
ALRTHAR2(200)
CATE SPNOS
NUMBER
RESERVATION
PK FK | NUMBER
MCEEDOE_ID AH AUDBBL
START_TIME DATE
END_TIME DATE
APPLICATIONS
PK ID | NUMBER
POST_ID VACHAR (NULLL
USER_ID VARCHAR2(300)
MESSAGE CCOOO
CREATED_AT DATE
POSTS
PK ID | NUMBER
POST_ID NUMBER
CREATED_AT DATE
SPACE
PK NUMBER
SPACE_ID BPK
NUMBER
SCHEDULE
PK SCHEDULE_ID DATEOR
START_TIME DATE
DATE
RESERVATION
PK RESERVATION_ID NUMBER
SPACE_ID VARCHAR2(200)
NUMBER_NUMBEER
SCHEDULE
PK SCHEDULE_ID
START_TIME
END_TIME
1:N
1:1
M:N
1:M
PK
FK

04 Relation Schema



05 사용자 DDL

```
CREATE TABLE USERS (
    ID              NUMBER          PRIMARY KEY,
    STUDENT_ID      VARCHAR2(20)    NOT NULL,
    NAME            VARCHAR2(50)    NOT NULL,
    MAJOR           VARCHAR2(100),
    PASSWORD_HASH   VARCHAR2(400)   NOT NULL,
    CREATED_AT      DATE,
    INTEREST        VARCHAR2(100)
);
```

05 활동 , 카테고리 DDL

```
CREATE TABLE ACTIVITIES (
    ID                NUMBER          PRIMARY KEY,
    TITLE             VARCHAR2(200)   NOT NULL,
    TAGS              VARCHAR2(400),
    DESCRIPTION       VARCHAR2(2000),
    LOCATION          VARCHAR2(200),
    START_TIME        DATE,
    END_TIME          DATE,
    MAX_PARTICIPANTS  NUMBER,
    CREATED_BY        NUMBER,
    CREATED_AT        DATE,
    CATEGORY_ID       NUMBER          NOT NULL
);
```

```
CREATE TABLE CATEGORIES (
    ID            NUMBER           PRIMARY KEY,
    NAME          VARCHAR2(400)    NOT NULL,
    DESCRIPTION   VARCHAR2(200),
    COLOR         VARCHAR2(30),
    CREATED_AT    DATE,
    PARENT_ID     NUMBER,
    ICON          NVARCHAR2(20)
);
```

05 게시글, 신청서 DDL


```
CREATE TABLE CATEGORY_POST (
    POST_ID       NUMBER          PRIMARY KEY,
    CATEGORY_ID   NUMBER          NOT NULL,
    USER_ID       VARCHAR2(20)    NOT NULL,
    TITLE         VARCHAR2(200)   NOT NULL,
    CONTENT       CLOB            NOT NULL,
    CREATED_AT    DATE,
    IMAGE_PATH    VARCHAR2(300)
);
```

```
CREATE TABLE APPLICATIONS (
    ID            NUMBER          PRIMARY KEY,
    POST_ID       NUMBER          NOT NULL,
    APPLICANT_ID  VARCHAR2(20)    NOT NULL,
    MESSAGE       VARCHAR2(2000),
    STATUS        VARCHAR2(20),
    CREATED_AT    DATE,
    UPDATED_AT    DATE
);
```

05 공간, 예약 DDL

```
CREATE TABLE "DB2025_501_7"."SPACE"
(
    "SPACE_ID" VARCHAR2(20),
    "SPACE_NAME" VARCHAR2(100) NOT NULL ENABLE,
    "TYPE" VARCHAR2(50),
    "LOCATION" VARCHAR2(255),
    "CAPACITY" NUMBER,
    "CURRENT_STATUS" VARCHAR2(50),
    "REQUIRES_APPROVAL" CHAR(1) DEFAULT 'Y',
    "AVAILABLE_START" VARCHAR2(5),
    "AVAILABLE_END" VARCHAR2(5),
    "MAX_DURATION" NUMBER,
    PRIMARY KEY ("SPACE_ID")
);
```

```
CREATE TABLE "DB2025_501_7"."RESERVATION"
(
    "RESERVE_ID" NUMBER,
    "STUDENT_ID" VARCHAR2(20) NOT NULL ENABLE,
    "SPACE_ID" VARCHAR2(20) NOT NULL ENABLE,
    "START_TIME" DATE NOT NULL ENABLE,
    "END_TIME" DATE NOT NULL ENABLE,
    "QR_AUTH_YN" CHAR(1) DEFAULT 'N',
    "VIOLATION_YN" CHAR(1) DEFAULT 'N',
    "STATUS" VARCHAR2(20) DEFAULT 'PENDING',
    "SEAT_NO" VARCHAR2(10),
    "REASON" VARCHAR2(200),
    CHECK (QR_Auth_Yn IN ('Y', 'N')) ENABLE,
    CHECK (Violation_Yn IN ('Y', 'N')) ENABLE,
    PRIMARY KEY ("RESERVE_ID")
);
```

05 일정 DDL

```
CREATE TABLE SCHEDULE (
    SCHEDULE_ID NUMBER PRIMARY KEY,
    STUDENT_ID VARCHAR2(20) NOT NULL,
    ACTIVITY_ID NUMBER,
    RESERVE_ID NUMBER,
    TYPE VARCHAR2(50) NOT NULL,
    START_TIME DATE NOT NULL,
    END_TIME DATE NOT NULL,
    NOTIFICATION_SETTING CHAR(1) DEFAULT 'N'
        CHECK (NOTIFICATION_SETTING IN ('Y','N')),
    TITLE VARCHAR2(255),
    CONSTRAINT FK_SCHED_RESERVATION
        FOREIGN KEY (RESERVE_ID)
        REFERENCES RESERVATION(RESERVE_ID)
);
```

05 커뮤니티, 댓글 DDL

```sql
CREATE TABLE BOARDS
(
    BOARD_ID    NUMBER PRIMARY KEY,
    CATEGORY_ID NUMBER NOT NULL,
    USER_ID     VARCHAR2(20) NOT NULL,
    TITLE       VARCHAR2(200) NOT NULL,
    CONTENT     CLOB NOT NULL,
    CREATED_AT  DATE DEFAULT SYSDATE NOT NULL,
    IS_ANON     CHAR(1) DEFAULT 'N',
    CHECK (IS_ANON IN ('Y','N'))
);
```

```sql
CREATE TABLE COMMENTS
(
    COMMENT_ID NUMBER PRIMARY KEY,
    BOARD_ID   NUMBER NOT NULL,
    USER_ID    VARCHAR2(20) NOT NULL,
    CONTENT    VARCHAR2(1000) NOT NULL,
    CREATED_AT DATE DEFAULT SYSDATE NOT NULL,
    IS_ANON    CHAR(1) DEFAULT 'N',
    CHECK (IS_ANON IN ('Y','N'))
);
```

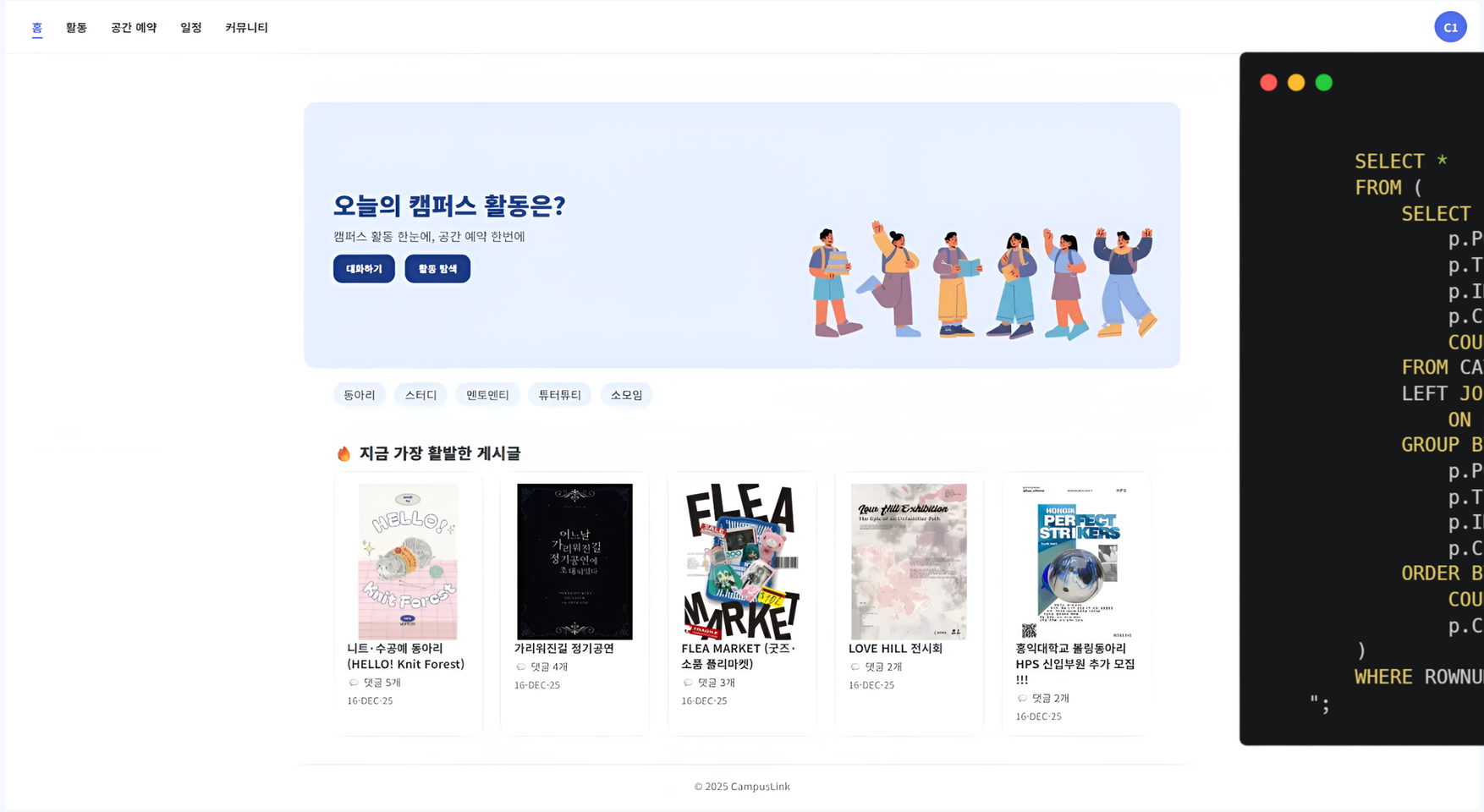
05 PHP & ORACLE 연결

db_conn.php

```php
<?php
header('Content-Type: text/html; charset=utf-8');
putenv("NLS_LANG=KOREAN_KOREA.AL32UTF8");

$host = "203.249.87.57";
$port = "1521";
$sid = "orcl";
$user = "DB2025_501_7";
$pw = "DB2025_501_7";

$tns = "(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCP)(HOST=$host)(PORT=$port)))(CONNECT_DATA=(SID=$sid)))";

$conn = oci_connect($user, $pw, $tns, 'AL32UTF8');

if (!$conn) {
    $e = oci_error();
    echo "<h2 style='color:red'>DB 연결 실패: " . $e['message'] . "</h2>";
    exit;
}
?>
```

06 웹 페이지 구현

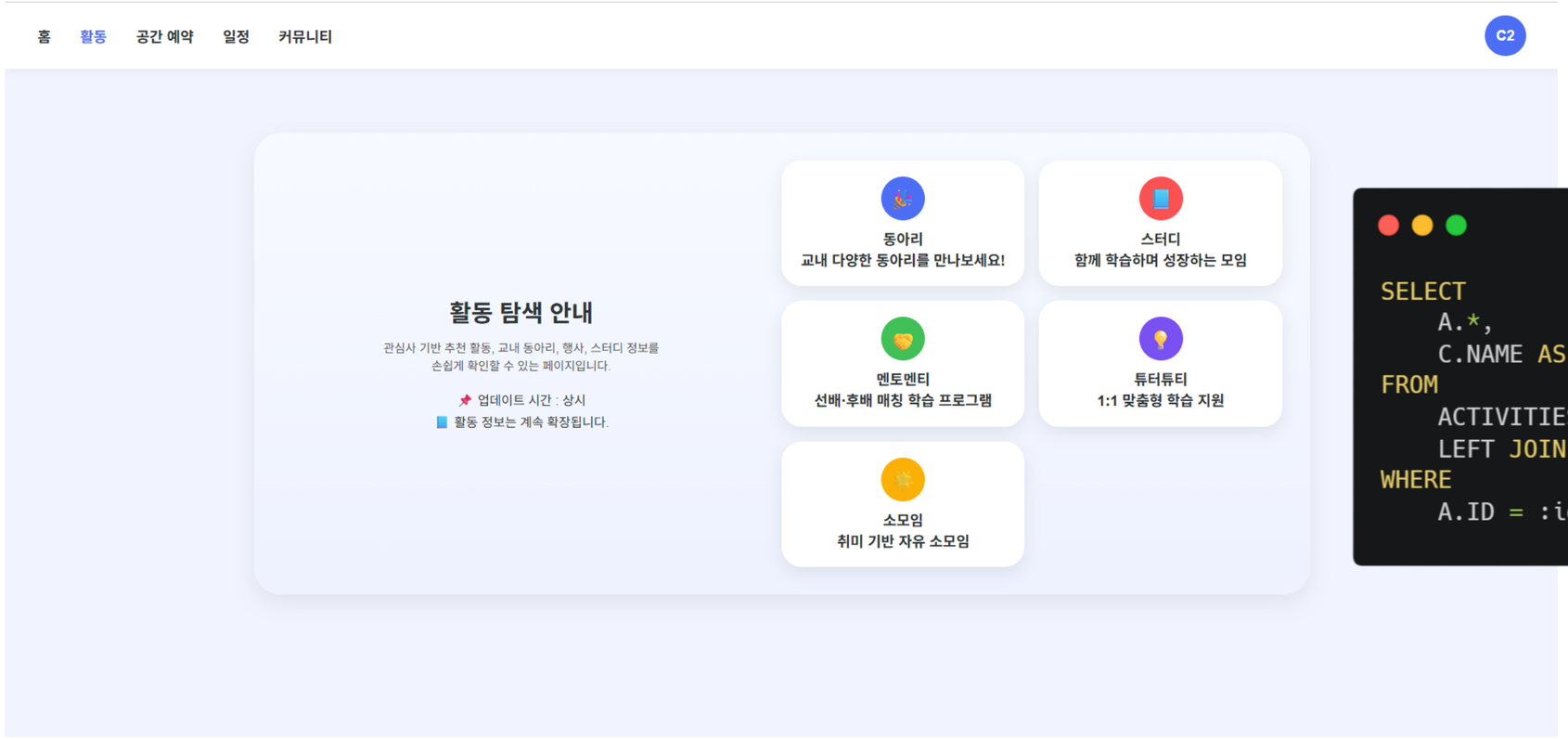
메인화면(index.php)




```
        SELECT *
        FROM (
            SELECT
                p.POST_ID,
                p.TITLE,
                p.IMAGE_PATH,
                p.CREATED_AT,
                COUNT(c.COMMENT_ID) AS COMMENT_COUNT
            FROM CATEGORY_POST p
            LEFT JOIN COMMENTS c
                ON p.POST_ID = c.POST_ID
            GROUP BY
                p.POST_ID,
                p.TITLE,
                p.IMAGE_PATH,
                p.CREATED_AT
            ORDER BY
                COUNT(c.COMMENT_ID) DESC,
                p.CREATED_AT DESC
        )
        WHERE ROWNUM <= :limit
    ";
```

06 웹 페이지 구현

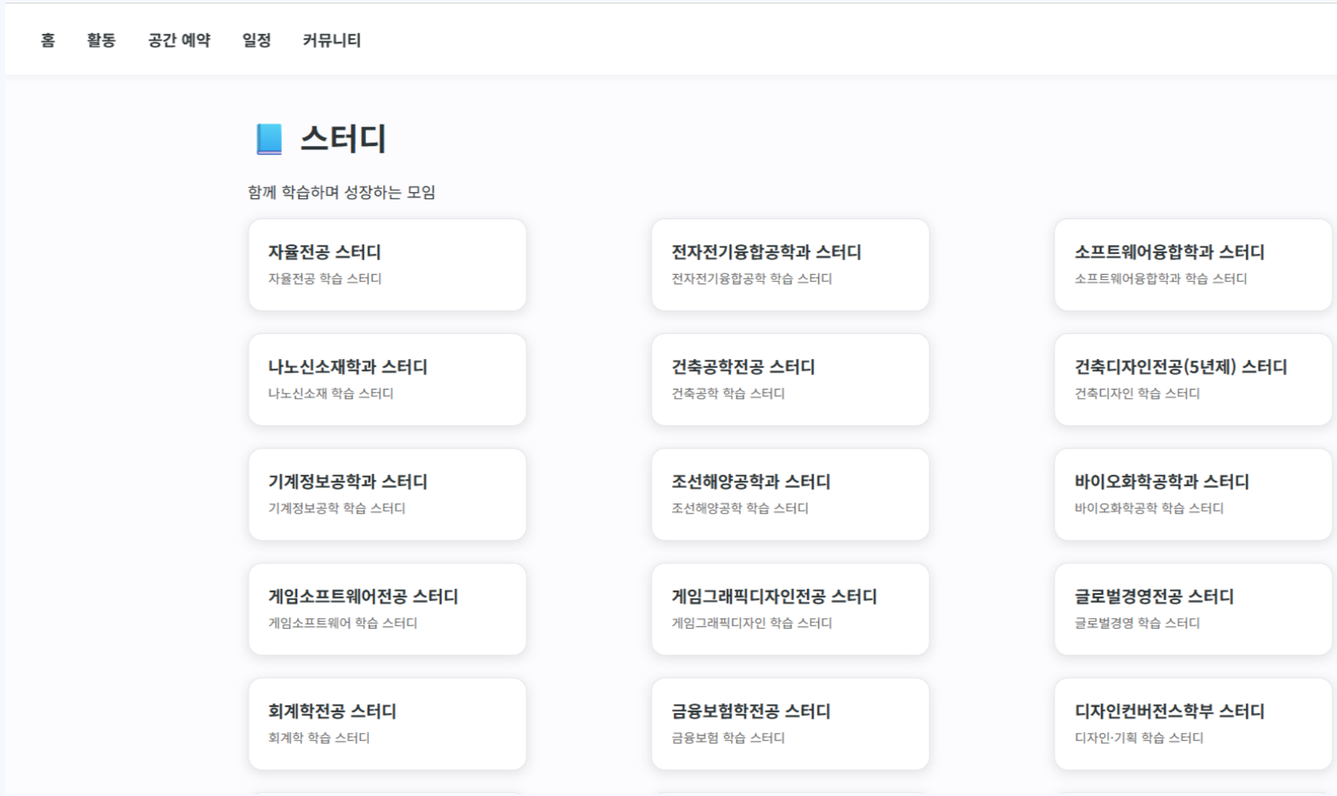
활동화면(activities.php)




```sql
SELECT
    A.*,
    C.NAME AS CATEGORY
FROM
    ACTIVITIES A
    LEFT JOIN CATEGORIES C ON A.CATEGORY_ID = C.ID
WHERE
    A.ID = :id
```

06 웹 페이지 구현

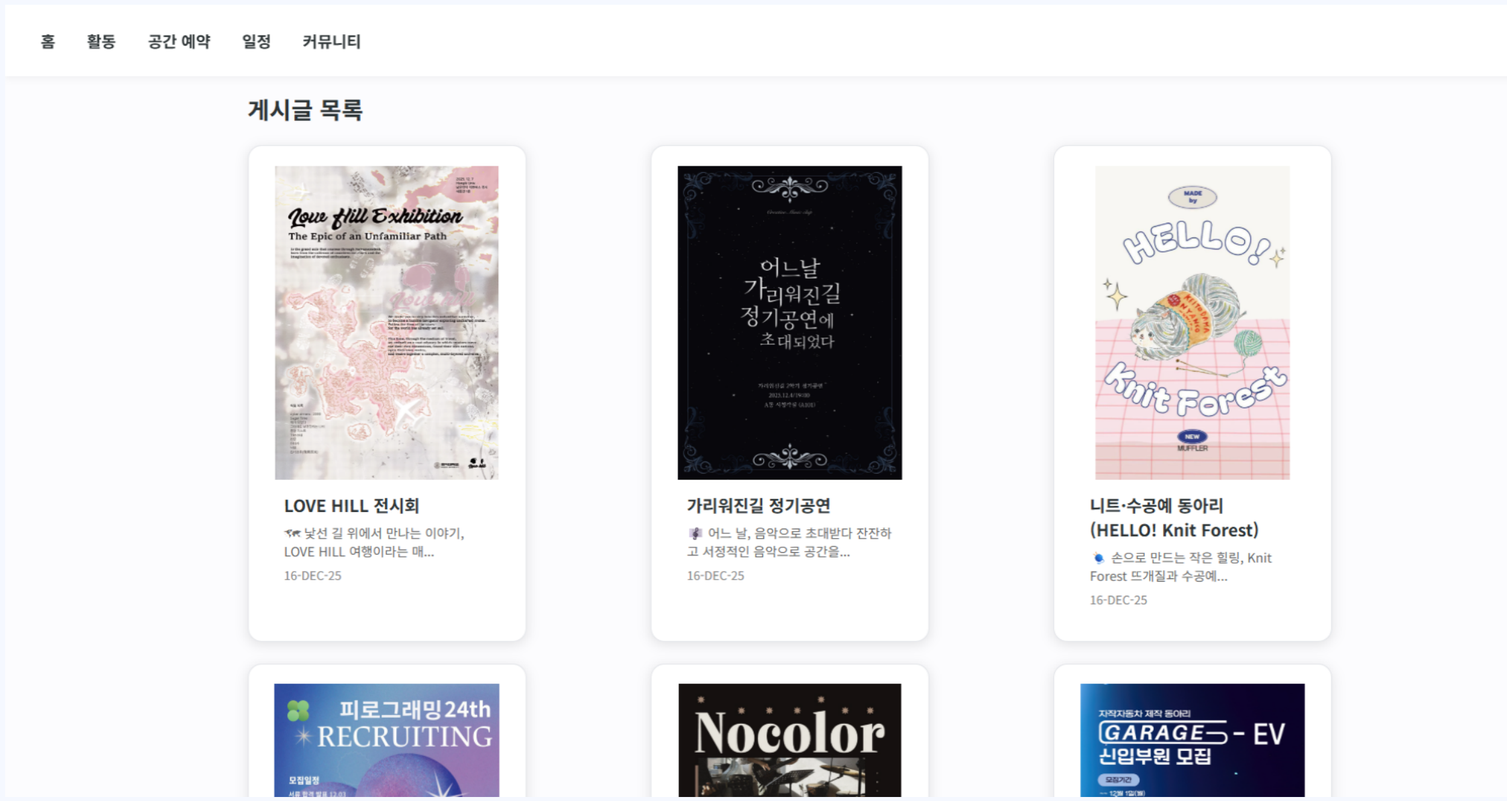
활동화면(category.php)




```sql
SELECT
    ID,
    NAME,
    DESCRIPTION,
    ICON,
    COLOR
FROM CATEGORIES
WHERE PARENT_ID = :id
ORDER BY ID ASC;
```

06 웹 페이지 구현

활동화면(category_view.php)




```sql
SELECT
    P.POST_ID,
    P.TITLE,
    P.CONTENT,
    P.CREATED_AT,
    P.IMAGE_PATH,
    P.USER_ID,
    U.NAME,
    P.CATEGORY_ID
FROM CATEGORY_POST P
JOIN USERS U
    ON P.USER_ID = U.STUDENT_ID
WHERE P.POST_ID = :post_id;
```

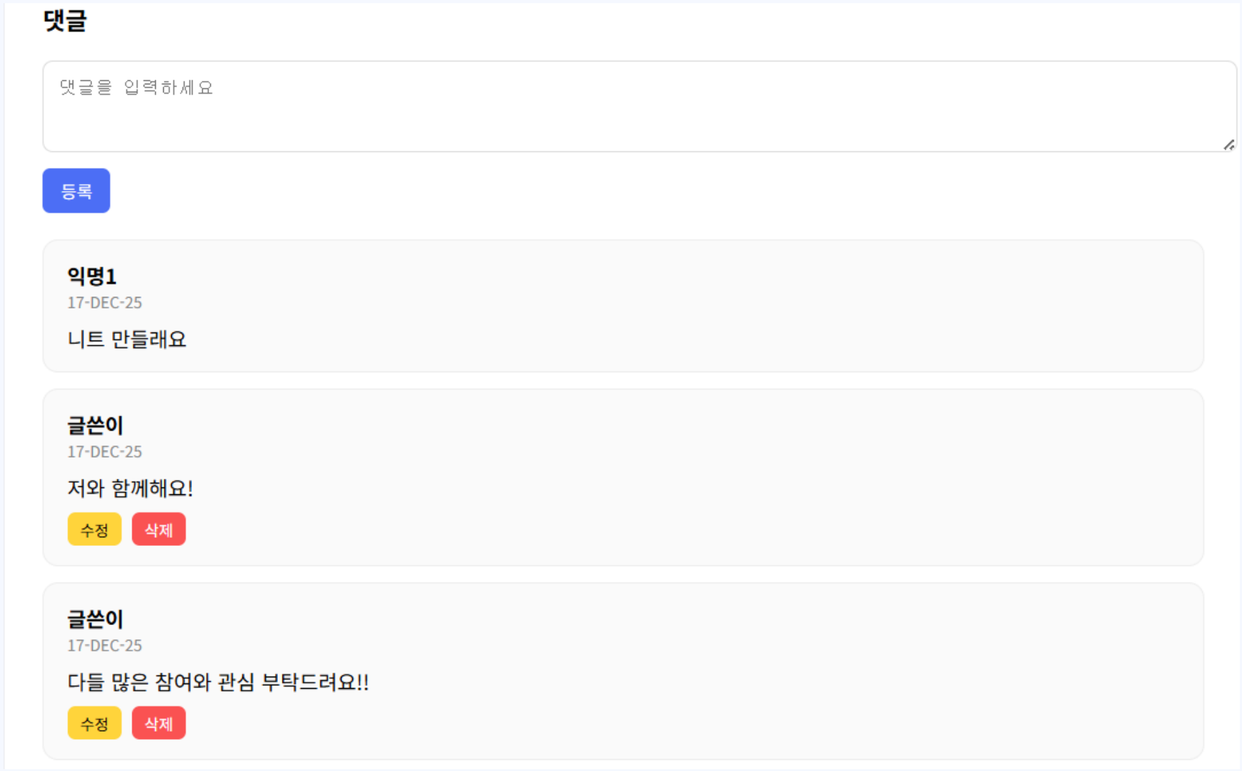
06 웹 페이지 구현

활동화면(category_view.php)

```sql
SELECT
    P.POST_ID,
    P.CATEGORY_ID,
    P.TITLE,
    P.CONTENT,
    P.IMAGE_PATH,
    P.USER_ID,
    P.CREATED_AT,
    U.NAME
FROM CATEGORY_POST P
JOIN USERS U
    ON P.USER_ID = U.STUDENT_ID
WHERE P.POST_ID = :pid;
```

06 웹 페이지 구현

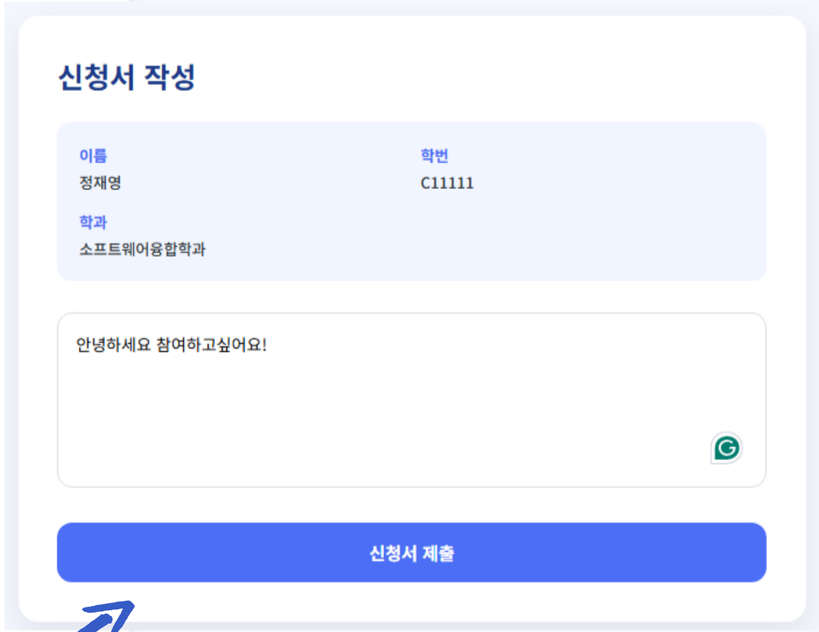
활동화면(category_view.php)




```sql
SELECT
    COMMENT_ID,
    POST_ID,
    USER_ID,
    CONTENT,
    CREATED_AT
FROM COMMENTS
WHERE POST_ID = :post_id
ORDER BY CREATED_AT ASC;
```

06 웹 페이지 구현

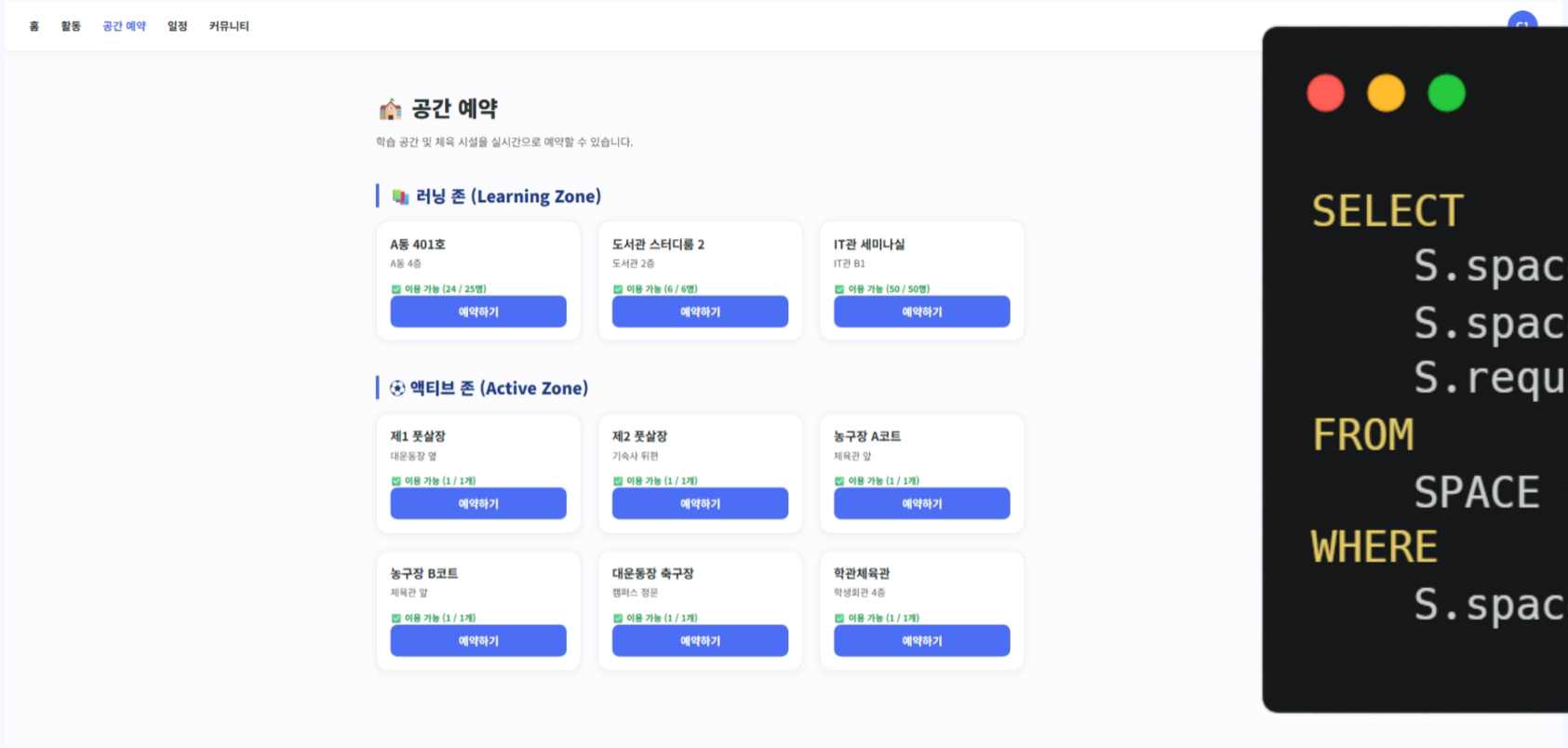
활동화면(apply_form.php)




```sql
SELECT
    a.ID                AS
APPLICATION_ID,
    a.MESSAGE,
    a.STATUS,
    a.CREATED_AT,

    p.POST_ID,
    p.TITLE,

    u.NAME,
    u.STUDENT_ID,
    u.MAJOR
FROM APPLICATIONS a
JOIN CATEGORY_POST p
    ON a.POST_ID = p.POST_ID
JOIN USERS u
    ON a.APPLICANT_ID = u.STUDENT_ID
WHERE a.APPLICANT_ID = :me
ORDER BY a.CREATED_AT DESC;
```

06 웹 페이지 구현

공간예약화면(reservation.php)




```sql
SELECT
    S.space_id,
    S.space_name,
    S.requires_approval
FROM
    SPACE S
WHERE
    S.space_id = :selected_id;
```

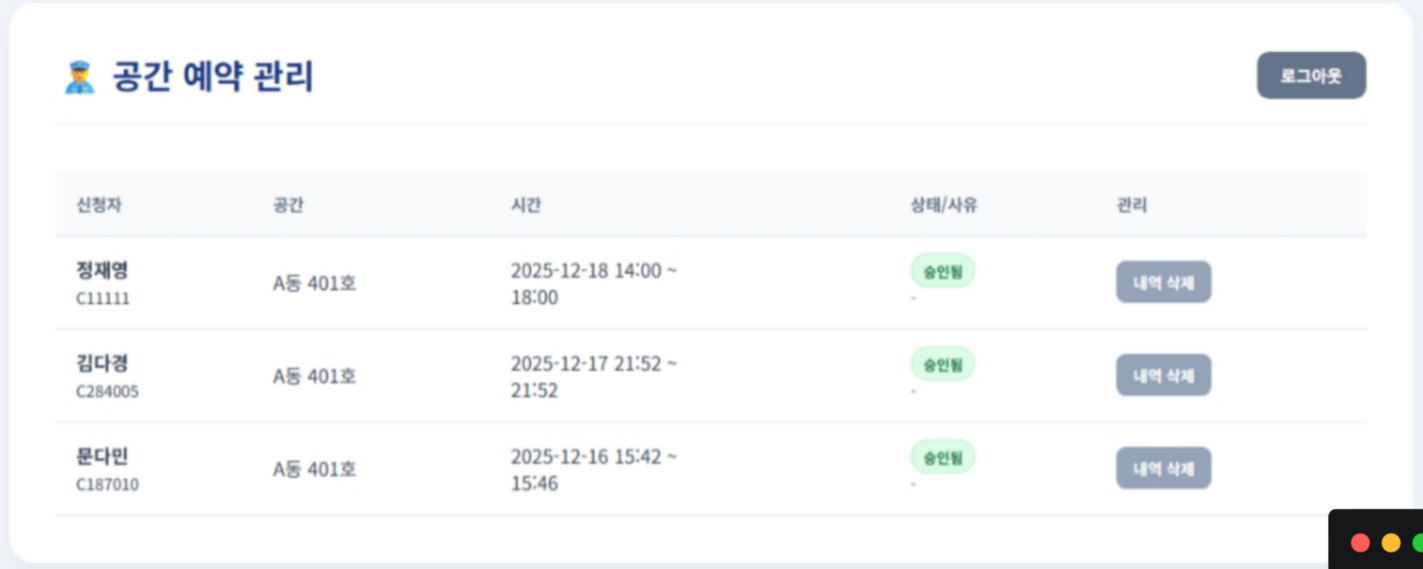
06 웹 페이지 구현

공간예약화면(reservation.php)

```
INSERT INTO
    RESERVATION
VALUES
    (
     (SELECT MAX(ID)+1 FROM RESERVATION),
     :student_id,
     :space_id,
     :start_time,
     :status
    );
```

06 웹 페이지 구현

공간예약관리화면(admin.php)

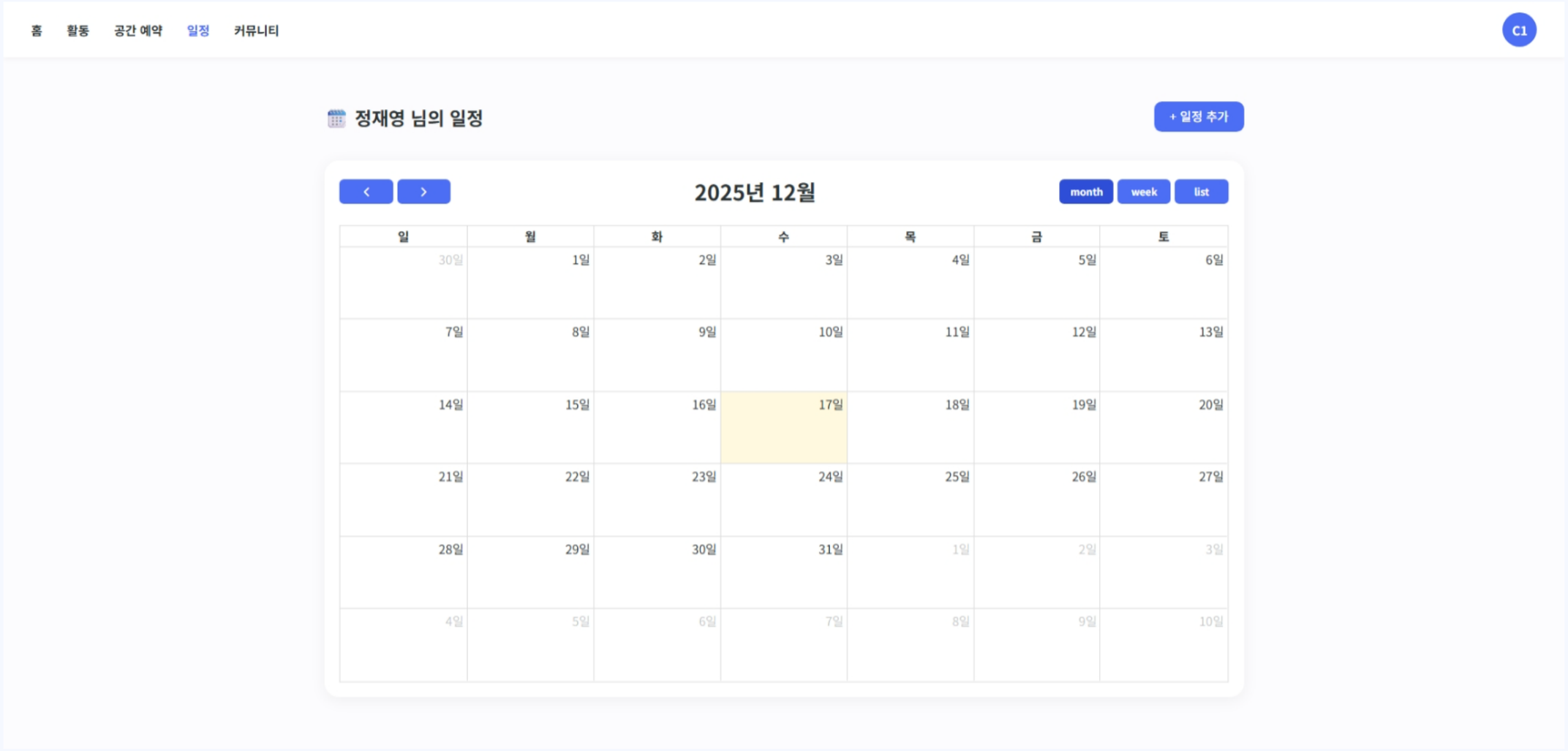



```sql
SELECT
    U.name,
    U.student_id,
    S.space_name,
    R.start_time,
    R.end_time,
    R.reason,
    R.reserve_id
FROM
    RESERVATION R
JOIN
    USERS U ON R.student_id = U.student_id
JOIN
    SPACE S ON R.space_id = S.space_id
WHERE
    R.status = 'PENDING' 회
ORDER BY
    R.start_time ASC;
```

```sql
UPDATE
    RESERVATION
SET
    status = 'APPROVED' -- 거절 시: 'REJECTED'
WHERE
    reserve_id = :clicked_id; -- 관리자가 클릭한 그 예약 건만 변경
```

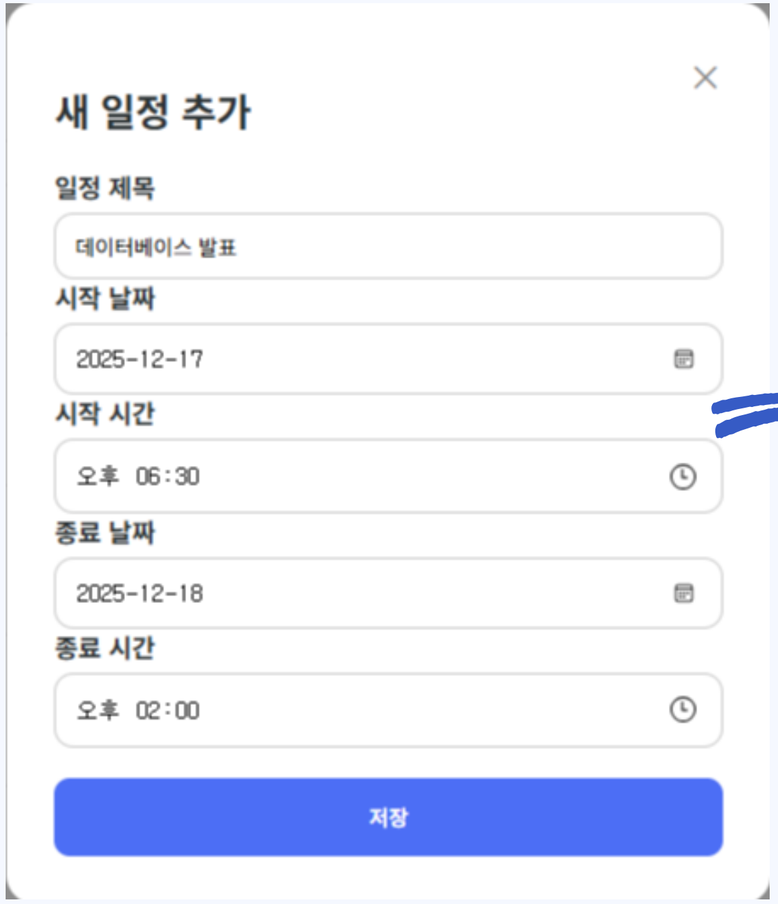
06 웹 페이지 구현

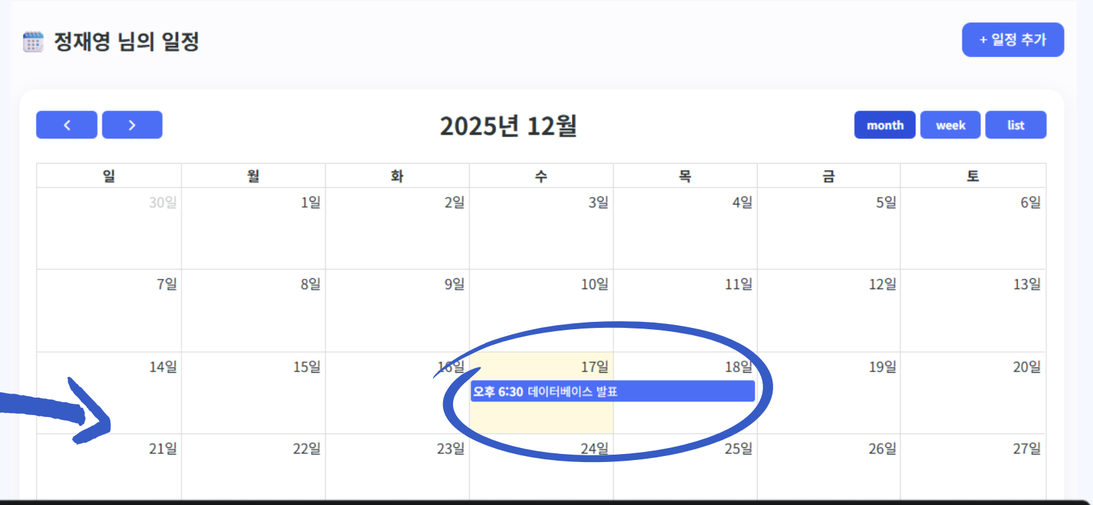
일정화면(calendar.php)



06 웹 페이지 구현

일정화면(calendar.php)

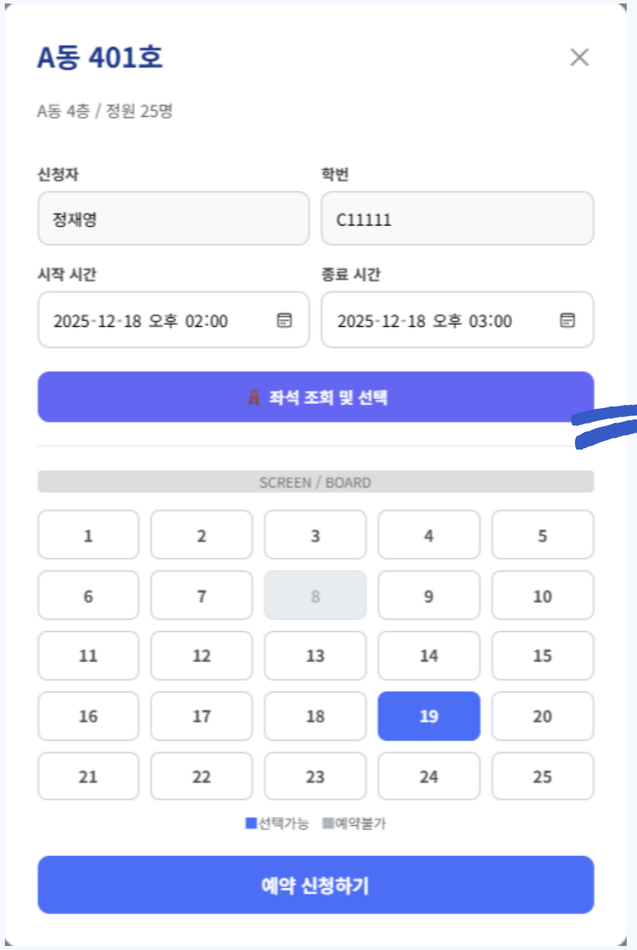


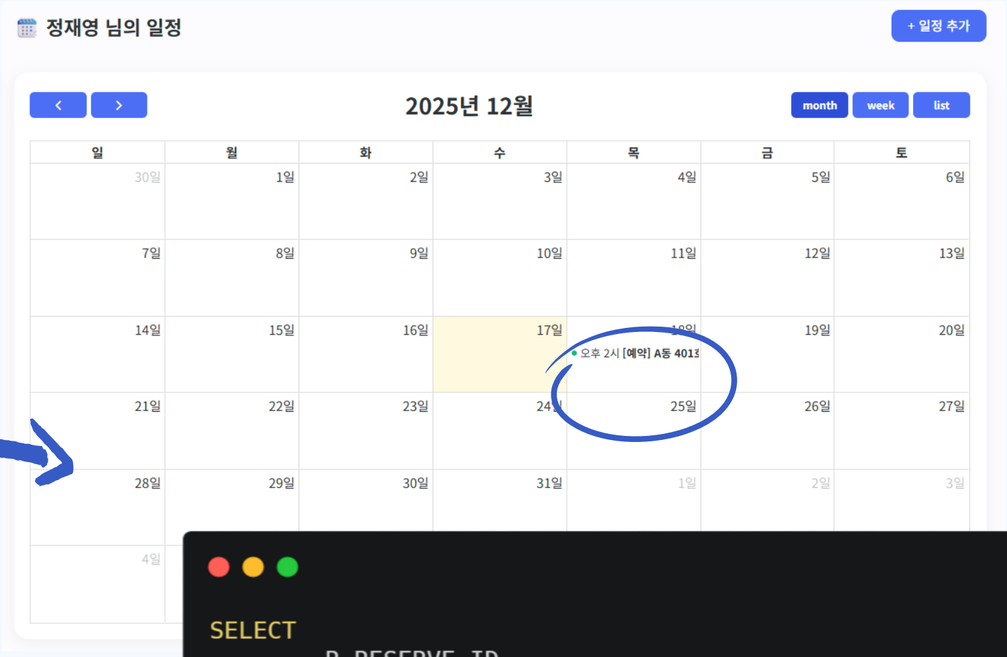



```
SELECT SCHEDULE_ID,
           TITLE,
           TO_CHAR(START_TIME, 'YYYY-MM-DD HH24:MI') AS START_TIME,
           TO_CHAR(END_TIME, 'YYYY-MM-DD HH24:MI') AS END_TIME
    FROM
        SCHEDULE
    WHERE
        UPPER(STUDENT_ID) = UPPER(:id)
```

06 웹 페이지 구현

일정화면(calendar.php)

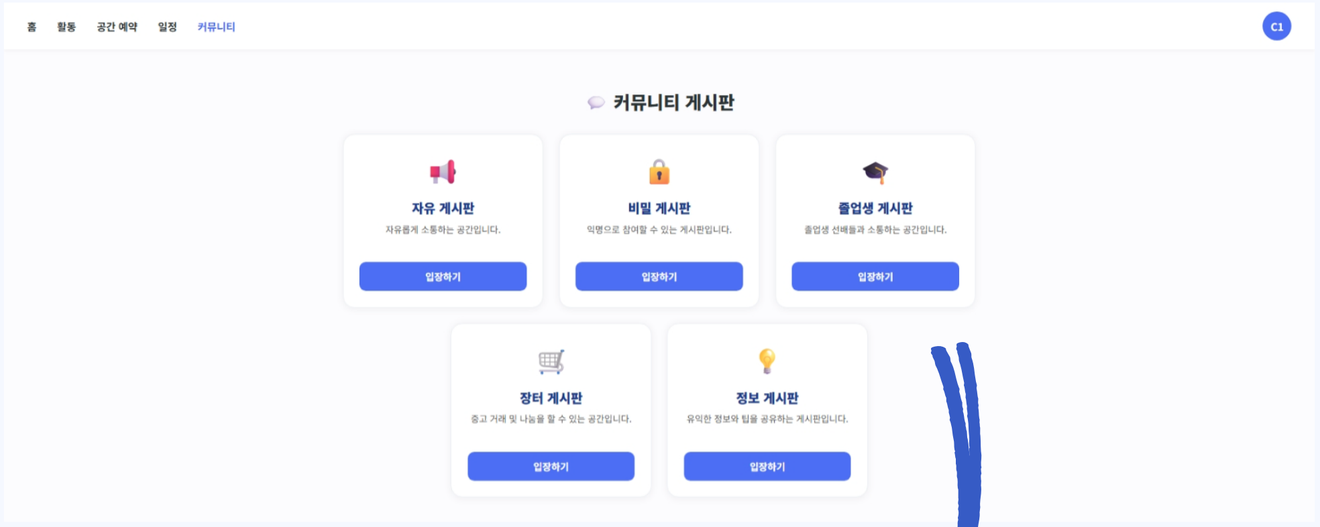


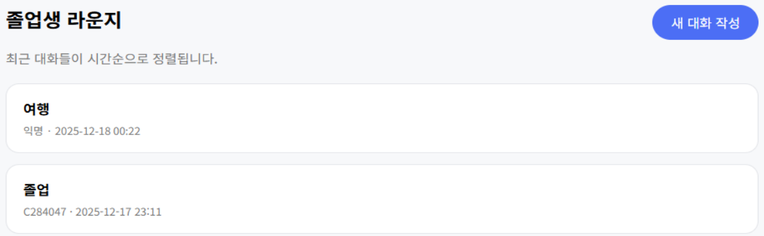



```oracle
SELECT
        R.RESERVE_ID,
        S.SPACE_NAME,
        TO_CHAR(R.START_TIME, 'YYYY-MM-DD HH24:MI') AS START_TIME,
        TO_CHAR(R.END_TIME, 'YYYY-MM-DD HH24:MI') AS END_TIME
    FROM
        RESERVATION R
    JOIN
        SPACE S
    ON
        R.SPACE_ID = S.SPACE_ID
    WHERE
        UPPER(R.STUDENT_ID) = UPPER(:id)
        AND R.STATUS = 'APPROVED'
```

06 웹 페이지 구현

커뮤니티화면(community.php-> board_list.php)

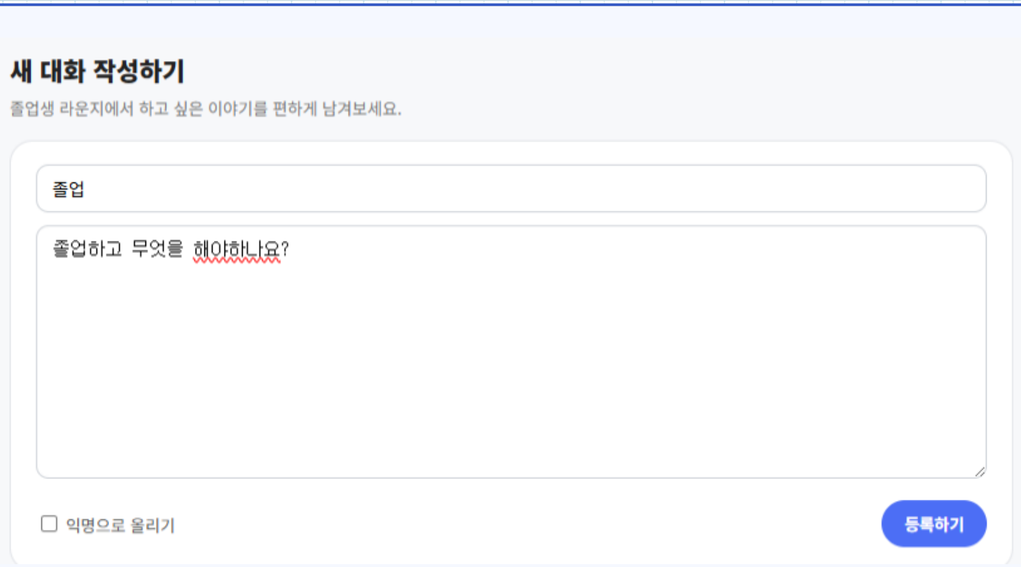


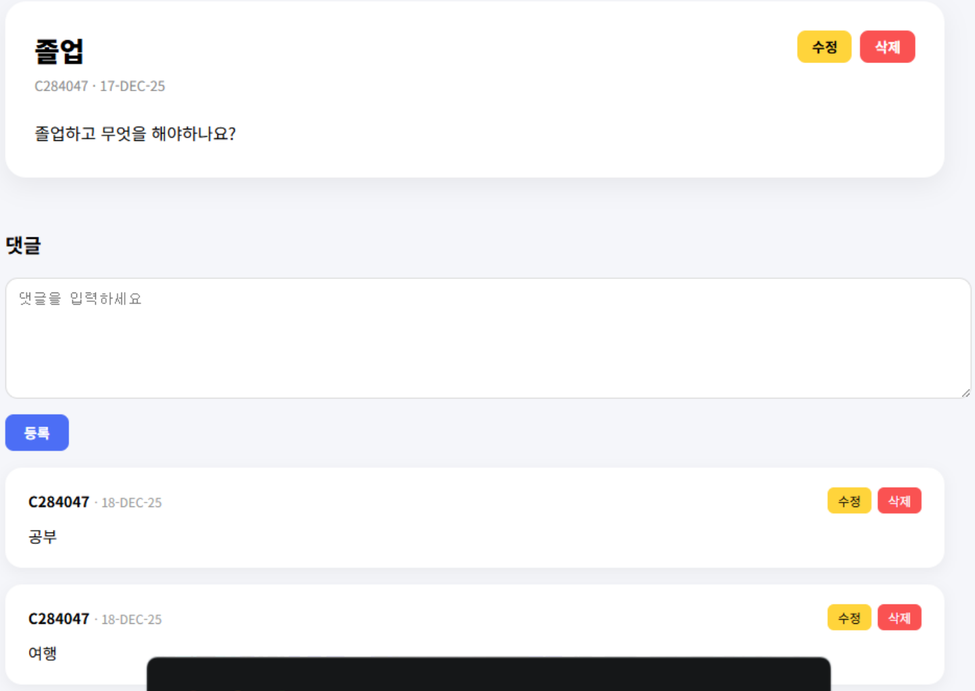



```sql
SELECT
    POST_ID, TITLE, CONTENT, AUTHOR,
   CREATED_AT
FROM
    POSTS
WHERE
    POST_ID = :pid;
```

06 웹 페이지 구현

커뮤니티화면(board_write.php)






```
SELECT
    COMMENT_ID, USER_ID, CONTENT, CREATED_AT
FROM
    COMMENTS
WHERE
    POST_ID = :pid
ORDER BY COMMENT_ID ASC;
```

```
SELECT
    COMMENT_ID, USER_ID, CONTENT
FROM
    COMMENTS
WHERE
    COMMENT_ID = :cid;
```